

Lecture 1 Exercise Solutions

Contents

1	Important Equations	2
2	Exercise 2.1.13	2
3	Exercise 2.1.22	3
4	Exercise 2.2.7	4
5	Exercise 2.2.10	5
6	Exercise 2.3.8	6
7	Exercise 2.4.2	7
8	Exercise 2.5.2	8
9	Exercise 2.5.7	9
10	Exercise 2.6.5	10
11	Exercise 2.7.4	11
12	Exercise 2.7.11	11
13	Exercise 2.7.12	12
14	Exercise 2.8.1	13

1 Important Equations

Baye's Theorem

Baye's theorem describes the probability of an event A given effects observed E_i :

$$P(E_i|A) = \frac{P(E_i \cap A)}{P(A)} = \frac{P(A|E_i)P(E_i)}{P(A|E_1)P(E_1) + \dots + P(A|E_n)P(E_n)} \quad (1)$$

Permutations (ordered sequences)

Permutations asks how many ways can n objects be arranged in k positions when order matters:

$$P_k^n = \frac{n!}{(n-k)!} \quad (2)$$

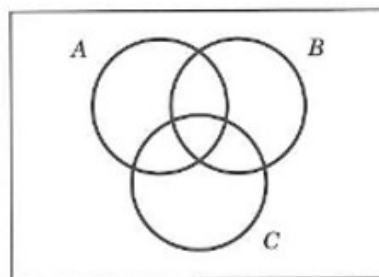
Combinations (unordered sequences)

Combinations asks in how many ways can k objects be selected from n objects when order does not matters:

$$C_k^n = \frac{P_k^n}{k!} = \frac{n!}{k!(n-k)!} \quad (3)$$

2 Exercise 2.1.13

Three events are shown in the Venn diagram in the following figure:



Reproduce the figure and shade the region that corresponds to each of the following events.

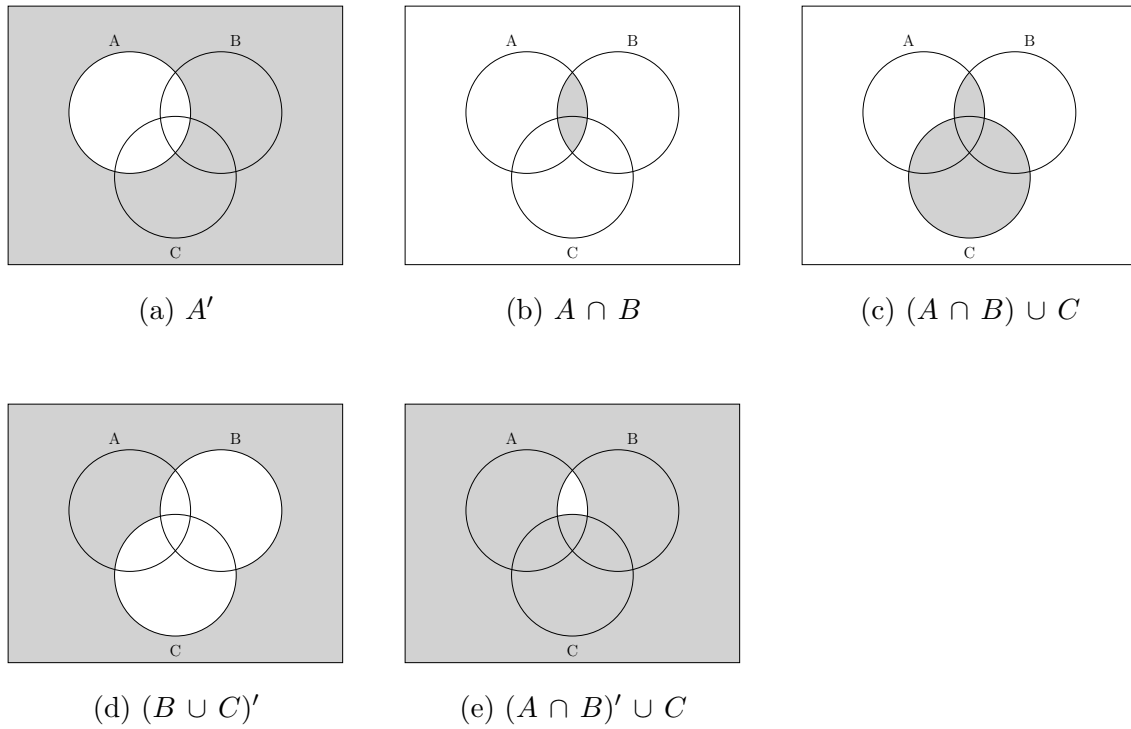


Figure 1: Solutions to exercise 2.1.13.

3 Exercise 2.1.22

The following table summarizes 204 endothermic reactions involving sodium bicarbonate.

Final Temperature Condition	Heat Absorbed [cal]	
	Below Target	Above Target
266 K	12	40
271 K	44	16
274 K	56	36

Let A denote the event that a reaction's final temperature is 271 K or less. Let B denote the event that the heat absorbed is below target. Determine the number of reactions in each of the following events.

a. $A \cap B$

This question ask what is the intersection between A and B i.e. how many reactions

had heat absorbtion below target at 271 K or less:

$$A \cap B = 44 + 12 = 56$$

b. A'

Since A denotes temperatures at 271 K or less we can add all reactions taking place at higher temperatures:

$$A' = 56 + 36 = 92$$

c. $A \cup B$

This is the union of A and B which means that all reactions that occur at 271 K or have heat absorption below the target should be added together:

$$A \cup B = 56 + 44 + 12 + 40 + 16 = 168$$

d. $A \cup B'$

This is almost the same as before however now B is negated. This means that all reactions that occur at 271 K or have heat absorption above the target should be added together:

$$A \cup B' = 44 + 12 + 40 + 16 + 36 = 148$$

e. $A' \cap B'$

For this we look at all reactions above 271 K and that has heat absorption above target:

$$A' \cap B' = 36$$

4 Exercise 2.2.7

A batch of 140 semiconductor chips is inspected by choosing a sample of 5 chips without replacement. Assume 10 of the chips do not conform to customer requirements.

a. How many different samples are possible?

Because the order of drawn chips is not important this is a combination problem

and is solved using eq: 3 with $n = 140$ and $k = 10$:

$$C_5^{140} = \frac{140!}{5!(140-5)!} = 416965528$$

b. How many samples contain exactly one nonconforming chip?

We need to choose a sample so that exactly one of the five chips is nonconforming and the rest works. This means that we need to pick 1 chip of the 10 nonconforming chips and 4 chips of the 130 working chips. Using the rule of product theorem:

$$\frac{10!}{1!(10-1)!} \cdot \frac{130!}{4!(130-4)!} = 113588800$$

c. How many samples contain at least one nonconforming chip?

First the total number of samples is used as found in a. Using this the number of samples where 5 chips is picked from the 130 working chips is subtracted. This guarantees that at least one chip is nonconforming in the sample:

$$\frac{140!}{5!(140-5)!} - \frac{130!}{5!(130-5)!} = 130721752$$

5 Exercise 2.2.10

In the layout of a printed circuit board for an electronic product, 12 different locations can accommodate chips.

a. If five different types of chips are to be placed on the board, how many different layouts are possible?

In this part order matters since the chips are different meaning it is permutations. Here $n = 12$ locations and $k = 5$ chips:

$$P_5^{12} = \frac{12!}{(12-5)!} = 95040$$

b. If the five chips that are placed on the board are of the same type, how many different layouts are possible.

Because the chips are the same, the order does not matter, which means that we can use combinations:

$$C_5^{12} = \frac{12!}{5!(12-5)!} = 792$$

6 Exercise 2.3.8

Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch Resistance	High	70	9
	Low	16	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. If a disk is selected at random, determine the following probabilities:

a. $P(A)$

Since event A contains all samples with high shock resistance the probability is:

$$P(A) = \frac{70 + 16}{100} = 0.86$$

b. $P(B)$

Event B is all the samples with high scratch resistance:

$$P(B) = \frac{70 + 9}{100} = 0.79$$

c. $P(A')$

This is the inverse problem of a.:

$$P(A') = 1 - P(A) = 1 - 0.86 = 0.14$$

d. $P(A \cap B)$

This problem asks what the probability of selecting a disk with both high scratch resistance and shock resistance:

$$P(A \cap B) = \frac{70}{100} = 0.7$$

e. $P(A \cup B)$

Here we should find the probability of the union of A and B meaning all values in both events should be added:

$$P(A \cup B) = \frac{70 + 9 + 16}{100} = 0.95$$

f. $P(A' \cup B)$

This is the same idea as before but now the union of everything that is not in A and what is in B:

$$P(A' \cup B) = \frac{70 + 9 + 5}{100} = 0.84$$

7 Exercise 2.4.2

If $P(A) = 0.3$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine the following probabilities:

a. $P(A')$

This probability is for everything outside of A meaning:

$$P(A') = 1 - P(A) = 1 - 0.3 = 0.7$$

b. $P(A \cup B)$

This can be solved using the probability of a union theorem:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.2 - 0.1 = 0.4$$

c. $P(A' \cap B)$

Following the total probability rule for two events (eq. 2.11 in the book):

$$P(A' \cap B) = P(B) - P(A \cap B) = 0.2 - 0.1 = 0.1$$

d. $P(A \cap B')$

The total probability rule is again used to calculate the probability:

$$P(A \cap B') = P(A) - P(A \cap B) = 0.3 - 0.1 = 0.2$$

e. $P[(A \cup B)']$

This is the inverse probability of b. meaning:

$$P[(A \cup B)'] = 1 - P(A \cup B) = 1 - 0.4 = 0.6$$

f. $P(A' \cup B)$

Using the probability of a union theorem gives:

$$P(A' \cup B) = P(A') + P(B) - P(A' \cap B) = 1 - P(A) + P(B) - P(A' \cap B) = 0.7 + 0.2 - 0.1 = 0.8$$

8 Exercise 2.5.2

Samples of skin experiencing desquamation are analyzed for both moisture and melanin content. The results from 100 skin samples are as follows:

		Melanin Content	
		High	Low
Moisture Content	High	13	7
	Low	48	32

Let A denote the event that a sample has low melanin content, and the B denote the event that a sample has high moisture content. Determine the following probabilities:

a. $P(A)$

The probability of low melanin is found by adding the two values of the moisture content event that has this and dividing by total amount of samples:

$$P(A) = \frac{7 + 32}{100} = 0.39$$

b. $P(B)$

The probability of high moisture content can be found by adding the two cases where this is the case and dividing by total amount of samples:

$$P(B) = \frac{13 + 7}{100} = 0.2$$

c. $P(A|B)$

This probability is calculated using the conditional probability theorem (eq. 2.9 in the book):

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{7}{20}$$

Since there is 7 samples with both low melanin content and high moisture content.

d. $P(B|A)$

The conditional probability theorem is again used to calculate the probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{7}{39}$$

9 Exercise 2.5.7

Suppose A and B are mutually exclusive events. Construct a Venn diagram that contains the three events A , B , and C such that $P(A|C) = 1$ and $P(B|C) = 0$.

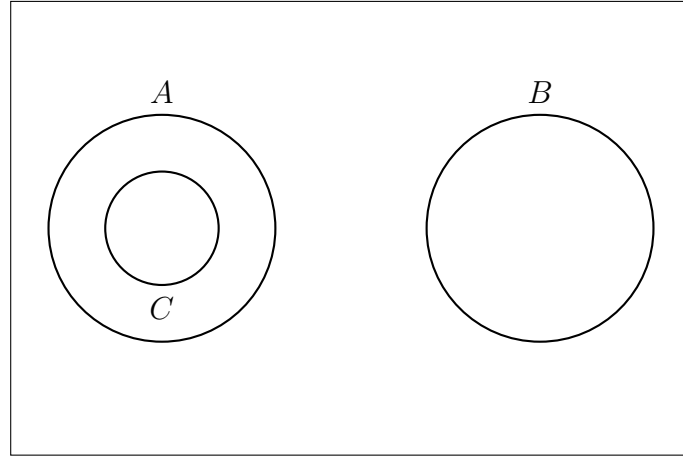


Figure 2: Venn diagram as solution to exercise 2.5.7.

The Venn diagram is shown in fig. 2. There should be 100% probability of A when C has happened and no probability of B when C has happened.

10 Exercise 2.6.5

The edge roughness of slit paper products increases as knife blades wear. Only 1% of products slit with new blades have rough edges, 3% of products slit with blades of average sharpness exhibit roughness, and 5% of products slit with worn blades exhibit roughness. If 25% of the blades in manufacturing are new, 60% are of average sharpness, and 15% are worn, what is the proportion of products that exhibit edge roughness?

The probability is calculated with the total probability rule using probabilities of a product exhibiting rough edges (re) with new blades (nb), average sharpness (as) and worn blades (wb):

$$\begin{aligned} P(re) &= P(re|nb) \cdot 1\% + P(re|as) \cdot 3\% + P(re|wb) \cdot 5\% \\ &= 0.25 \cdot 0.01 + 0.6 \cdot 0.03 + 0.15 \cdot 0.05 = 2.8\% \end{aligned}$$

Meaning the probability of a product exhibiting rough edges is 2.8%.

11 Exercise 2.7.4

Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch Resistance	High	70	9
	Low	16	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. Are events A and B independent?

If A and B is independent then (page 37, eq. 2.13):

$$P(A \cap B) = P(A)P(B)$$

Both sides are calculated using that the total amount of disks are 100:

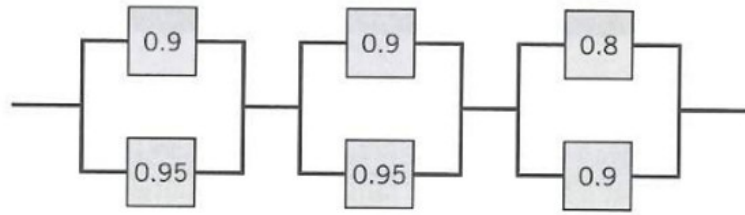
$$P(A)P(B) = \frac{70 + 16}{100} \cdot \frac{70 + 9}{100} = 0.6794$$

$$P(A \cap B) = \frac{70}{100} = 0.7$$

This means that events A and B are not independent since $P(A \cap B) \neq P(A)P(B)$.

12 Exercise 2.7.11

The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device is functional does not depend on whether or not other devices are functional. What is the probability that the circuit operates?



Using example 2.23 and 2.24 in the book the probability of an isolated parallel connection either working in the top T or bottom B is:

$$P(T \text{ or } B) = 1 - P[(T \text{ or } B)'] = 1 - P(T' \text{ and } B') = 1 - P(T')P(B')$$

First the probability is calculated in each of the parallel connections:

$$P_1 = 1 - (1 - 0.9)(1 - 0.95) = 0.995$$

$$P_2 = 1 - (1 - 0.9)(1 - 0.95) = 0.995$$

$$P_3 = 1 - (1 - 0.8)(1 - 0.9) = 0.98$$

The total probability of the circuit working is then using serial connections from example 2.23:

$$P = P_1 \cdot P_2 \cdot P_3 = 0.995 \cdot 0.995 \cdot 0.98 = 0.97$$

13 Exercise 2.7.12

The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device is functional does not depend on whether or not other devices are functional. What is the probability that the circuit operates?



Using the same idea as before, first the serial connections are calculated for the top and bottom:

$$P(T) = 0.9 \cdot 0.8 \cdot 0.7 = 0.504$$

$$P(B) = 0.95 \cdot 0.95 \cdot 0.95 = 0.857$$

Now the total probability of the device working is calculated as:

$$P = 1 - (1 - 0.504)(1 - 0.857) = 0.929$$

14 Exercise 2.8.1

Customers are used to evaluate preliminary product design. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor products.

a. What is the probability that a products attains a good review?

Using the total probability rule (page 36, eq. 2.12) the probability of a good review is:

$$P(\text{good review}) = 0.95 \cdot 0.4 + 0.6 \cdot 0.35 + 0.1 \cdot 0.25 = 0.615$$

b. if a new design attains a good review, what is the probability that it will be a highly successful product?

This can be found using Baye's theorem:

$$\begin{aligned} P(\text{successful product}|\text{good review}) &= \frac{P(\text{good review}|\text{successful product})P(\text{successful product})}{P(\text{good review})} \\ &= \frac{0.95 \cdot 0.4}{0.615} = 0.618 \end{aligned}$$

c. If a product does not attain a good review, what is the probability that it will be a highly successful product?

This problem can again be solved using Baye's theorem as before but now negating the good review:

$$\begin{aligned} P(\text{successful product}|\text{good review}') &= \frac{P(\text{good review}'|\text{successful product})P(\text{successful product})}{P(\text{good review}')} \\ &= \frac{(1 - 0.95) \cdot 0.4}{1 - 0.615} = 0.052 \end{aligned}$$